UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Convert -20.125 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

QUESTION A2 **(5 marks)**

Question 1 is based on the following statement.

Mr. X is known as Father of Computer. He developed Analytical Machine in 1837 which has four main components, namely; memory, the mill, punch card reader and the punch and printed section. Analytical Machine is the successor of Y machine. Y machine is used to compute tables of numbers for naval navigation.

a) Name the inventor of Mr. X.

b) Identify the name of Y machine.

c) What is the function of Analytical Machine?

d) Map the components Analytical Machine to Von Neumann architecture

QUESTION A3 **(5 marks)**

a) Give ONE (1) characteristic of Symmetric Multiprocessor (SMP).

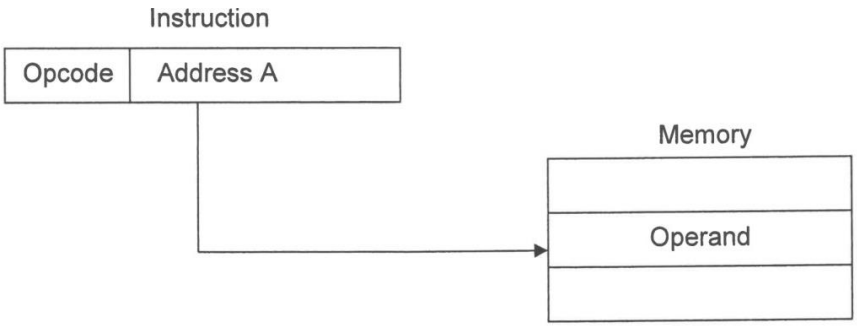
b) Discuss the meaning of "symmetric" in Symmetric Multiprocessor.

c) Give ONE (1) advantage of Symmetric Multiprocessor organization.

QUESTION A4 **(5 marks)**

Given the following Figure 2.

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a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A5 **(10 marks)**

a) Describe the function of each computer bus type.

b) Discuss TWO (2) benefits of synchronous bus timing.

QUESTION A6 **(10 marks)**

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: 24 - (-83)

a) Calculate 24 - (-83) using sign and magnitude.

b) Calculate 24 - (-83) using 1's and 2's complement.

QUESTION A7 **(10 marks)**

Processor contains a number of registers used for temporary storage of data. Describe FIVE (5) types of registers.

QUESTION A8 **(10 marks)**

a) Define interrupt.

b) Differentiate between interrupt and direct manipulation access.

c) Given the following scenario. Dilan is surfing woolworth.com to buy groceries online.

While he is comparing prices of products in woolworth.com and one other website, two messages popped up onto his screen. Figure 2 (a) and (b) illustrate the conveyed messages.

Are you sure you want to There is a hardware event failure in this computer. Please reboot cancel this request? this computer now.

(a)

(b)

i. Identify the type of interrupt in Figure 2(a) and Figure 2(b) with proper justification.

ii. While interruptroutinefor Interrupt Ais executing, it is interrupted by interrupt Bwhich has a higher priority than interrupt A. What will happen to interrupt routine A?

QUESTION A9 **(10 marks)**

a) Why input/output problem occurs?

b) Describe THREE (3) categories for input/output module and provide an appropriate example each.

QUESTION A10 **(10 marks)**

a) Rank the following memory types based on price ranging from the cheapest to the most expensive. Cache, tape back-up, register, hard drive

b) Discuss ONE (1) benefit of cache memory.

c) Differentiate between write through and write back mechanisms in order to store data in cache.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		